

# LIVE SYSTEM TRIAGE & VOLATILE ARTIFACT COLLECTION CHEAT SHEET

**Document Control ID:** DFIR-REF-003 **Target Standard:** RFC 3227 (Guidelines for Evidence Collection and Archiving) **Classification:** TECHNICAL REFERENCE // FIELD GUIDE

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## 1. RFC 3227 ORDER OF VOLATILITY EXECUTION MODEL

When acquiring evidence from a live system, collect data in order of decreasing volatility to prevent overwriting critical transient state artifacts:

1. **Registers & CPU Cache / Physical RAM** (Highest volatility - lost on power off)
2. **Routing Tables, ARP Cache, Process Tables, Kernel Memory**
3. **Active Network Sockets & Ephemeral Connections**
4. **Temporary File Systems, Swap Space / Pagefile**
5. **Persistent Storage & Disk Logs**
6. **Remote Logging & Archival Backups** (Lowest volatility)

**CORE FORENSIC RULE:** Run all triage binaries directly from an external, forensically prepared triage drive (E:\). Do NOT install software on the suspect host. Pipe all CLI outputs directly to the triage drive to prevent disk contamination.

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## 2. VOLATILE MEMORY (RAM) EXTRACTION COMMANDS

- **WinPmem (Windows Raw RAM Dump):** winpmem.exe -o E:\Triage\_Output\[CASE\_ID]\_[HOSTNAME]\_RAM.raw
  - **DumpIt (Automated Memory Capture):** DumpIt.exe /OUTPUT E:\Triage\_Output\[CASE\_ID]\_[HOSTNAME]\_RAM.dmp
  - **LiME (Linux Kernel Module Memory Capture):** insmod lime.ko "path=/mnt/triage\_usb/[CASE\_ID]\_[HOSTNAME]\_RAM.lime format=raw"
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## 3. LIVE WINDOWS TRIAGE CLI MATRIX

Run these commands from an elevated command prompt residing on your external triage volume (E:\):

| Target Artifact                     | Native Command Syntax                                                                                                        | Forensic Objective / Output                                                                              |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>System Time &amp; Date</b>       | <code>net time \\127.0.0.1 /Get-Date -Format o</code>                                                                        | Records exact clock skew against NIST/GPS reference time.                                                |
| <b>Active Network Sockets</b>       | <code>netstat -ano -b &gt; E:\Triage\network_sockets.txt</code>                                                              | Identifies listening ports, remote C2 connections, and associated Process IDs (PID).                     |
| <b>Routing Table &amp; ARP</b>      | <code>route print &amp; arp -a &gt; E:\Triage\routing_arp.txt</code>                                                         | Captures gateway redirects, MITM caches, and local subnet neighbors.                                     |
| <b>Process Tree &amp; Arguments</b> | <code>wmic process get ProcessId,ParentProcessId,CommandLine,ExecutablePath /format:csv &gt; E:\Triage\processes.csv</code>  | Full process tree and command line strings (detects encoded PowerShell, injected processes, and LOLBAS). |
| <b>Active Services</b>              | <code>Get-CimInstance Win32_Service   Select-Object Name,State,StartMode,PathName   Export-Csv E:\Triage\services.csv</code> | Identifies malicious persistence mechanisms masquerading as legitimate services.                         |
| <b>Scheduled Tasks</b>              | <code>schtasks /query /fo CSV /v &gt; E:\Triage\scheduled_tasks.csv</code>                                                   | Captures persistent scheduled tasks, triggers, and hidden script paths.                                  |
| <b>Logged-in Sessions</b>           | <code>query user &amp; qwinsta &gt; E:\Triage\user_sessions.txt</code>                                                       | Identifies active console, RDP, and disconnected background sessions.                                    |
| <b>Network Shares</b>               | <code>net share &amp; net use &gt; E:\Triage\smb_shares.txt</code>                                                           | Discovers mounted network drives and internal staging shares.                                            |
| <b>Loaded Drivers</b>               | <code>driverquery /v /fo CSV &gt; E:\Triage\drivers.csv</code>                                                               | Identifies unsigned, obsolete, or malicious kernel drivers (BYOVD exploits).                             |
| <b>Firewall Status</b>              | <code>netsh advfirewall show allprofiles &gt; E:\Triage\firewall_rules.txt</code>                                            | Verifies whether local perimeter firewall policies were disabled or modified.                            |

#### 4. POWER-STATE DECISION MATRIX

- **If System is Powered OFF:**
    - Keep powered OFF.
    - Remove AC power cable and battery (if applicable).
    - Transport inside an RF-shielded Faraday container.
    - Acquire via write-blocked dead-box disk imaging.
  - **If System is Powered ON and Disk is Encrypted (BitLocker / FileVault / LUKS):**
    - **DO NOT PULL THE PLUG.**
    - Capture RAM first (preserves in-memory volume decryption keys).
    - Extract volume recovery keys/escrow passwords.
    - Perform live logical or physical capture while volume is mounted and unlocked.
  - **If System is Powered ON and Active Malware is Wiping/Encrypting Disk:**
    - Immediately sever network connections (pull Ethernet / toggle physical Wi-Fi switch).
    - Immediately pull power cable from the rear of the machine to halt destructive write cycles.
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#### 5. TRIAGE LOG & MD5/SHA-256 INTEGRITY MANIFEST

Immediately after completing live triage commands, calculate cryptographic hashes for all collected output files:

```
Get-ChildItem -Path "E:\Triage_Output" -Recurse -File | Get-FileHash -Algorithm  
SHA256 | Export-Csv "E:\Triage_Output\Manifest_SHA256.csv" -NoTypeInformation
```

**Field Examiner Signature:** \_\_\_\_\_

**Printed Name:** [INSERT\_EXAMINER\_NAME], [CREDENTIALS]

**Date & Time Completed (UTC):** [INSERT\_TIMESTAMP\_UTC]